Sets

- A special kind of set, denoted by Σ^*.

Some definitions:

- **Alphabet**: to be a finite nonempty set Σ whose members are symbols, often called **letters** of Σ
- **Word**: A word is a finite string of letters from Σ. We denote the set of all words using letters from Σ by Σ^*.
- **Language**: Any subset of Σ^* is a called a language over Σ.

Ex: Let $\Sigma = \{a, b, c, \ldots, z\}$ consist of 26 letters of English alphabet. Then, Σ^* contains math, is, fun, ain't, fuzzyomph, etcetera, etc. Since Σ^* contains a, aa, aaa, aaaa, Σ^* is clearly an infinite set. Then, the English language is a subset of Σ^*, but a finite set.

- There is also an **empty word** λ analogous to the empty set.

Ex: Let $\Sigma = \{a, b\}$

then $\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, \ldots\}$.

Ex: Show that $f: \mathbb{N} \to \mathbb{N}$, $f(n) = 2n$ is 1-1.

Let $n_1, n_2 \in \mathbb{N}$ and $f(n_1) = f(n_2)$. Let us see that $n_1 = n_2$.

If $f(n_1) = f(n_2)$ then $2n_1 = 2n_2$. So $n_1 = n_2$. ✓

How to show 1-1:

If $x_1, x_2 \in S$ and $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$

or

If $x_1, x_2 \in S$ and $f(x_1) = f(x_2)$, then $x_1 = x_2$.

Is f onto?

Consider 3. There does not exist $n \in \mathbb{N}$ $f(n) = 3$

i.e $2n = 3$. Therefore it is not onto.

No, since $\text{Im}(f)$ consists of only of the even numbers.

Ex: Show that $f: \mathbb{R} \to \mathbb{R}$, $f(x) = 3x - 5$ is one-to-one correspondence.

For 1-1 we must show that for $x_1, x_2 \in \mathbb{R}$

If $f(x_1) = f(x_2)$, then $x_1 = x_2$.

If $f(x_1) = f(x_2)$, then $3x_1 - 5 = 3x_2 - 5 \Rightarrow 3x_1 = 3x_2$

$\Rightarrow x_1 = x_2$. ✓

For onto we must show that for each $y \in \mathbb{R}$ there is at least some $x \in \mathbb{R}$ s.t $f(x) = y$.

Let $f(x) = y$. I have to show that $x \in \mathbb{R}$. If $f(x) = y$, then $3x - 5 = y$. Then $3x = y + 5 \Rightarrow x = \frac{y+5}{3}$. Thus, if $y \in \mathbb{R}$, then $\frac{y+5}{3}$ is also in $\mathbb{R}$. So, for every y in $\mathbb{R}$, there is an $x \in \mathbb{R}$ s.t $f(x) = y$. ✓

Hence, f is one-to-one correspondence.

Characteristic Function.

Consider a set S and a subset A of S.

The function on S that takes the value 1 at the members of A and the value 0 at the other members of S is called **the characteristic function of A** and is denoted by χ_A. Thus $\chi_A(x) = \begin{cases} 1 & \text{for } x \in A \\ 0 & \text{for } x \in S - A \end{cases}$

Ex:
$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \quad \text{for } x \in \mathbb{R}.$$

$f(x)$ is the characteristic function $\chi_{[0,\infty)}$ defined on $\mathbb{R}$.

Ex: For each $n \in \mathbb{N}$, let $g(n)$ be the remainder when n is divided by 2.

Then $g: \mathbb{N} \to \{0, 1\}$

In fact, observe that g is the characteristic function of the set $A = \{n \in \mathbb{N} : n \text{ is odd}\}$, $g = \chi_A$.

Identity function:

The identity function 1_S on S is the function that maps each element of S to itself. **$1_S(x) = x$** for all $x \in S$. $(f(x) = x)$

Composition:

Consider the functions $f: S \to T$ and $g: T \to U$ we define the composition $g \circ f: S \to U$ by the rule

$$g \circ f(x) = g(f(x)) \text{ for all } x \in S.$$

Associativity of composition:

Consider the functions $f: S \to T$, $g: T \to U$, $h: U \to V$

Then $(h \circ g) \circ f = h \circ (g \circ f)$

Inverses of Functions:

An inverse of a function $f: S \to T$ is a function $f^{-1}: T \to S$ s.t. $f^{-1} \circ f = 1_S$ and $f \circ f^{-1} = 1_T$, i.e

s.t $f^{-1}(f(x)) = x$ for all $x \in S$.

$f(f^{-1}(y)) = y$ for all $y \in T$.

Theorem: The function $f: S \to T$ is invertible if and only if f is one-to-one and maps onto T.

Ex: Let $f: \mathbb{R} \to \mathbb{R}$ $f(x) = x^3 + 1$. Show that f is invertible.

For 1-1, we must show that if $f(x_1) = f(x_2)$ then $x_1 = x_2$. for $x_1, x_2 \in \mathbb{R}$

(if $f(x_1) = f(x_2)$, then $x_1^3 + 1 = x_2^3 + 1 \Rightarrow x_1^3 = x_2^3 \Rightarrow x_1 = x_2$.

because each real number has a unique cube root. ✓

For onto, consider $y \in \mathbb{R}$, we have to find $x \in \mathbb{R}$ s.t $f(x) = y$.

$x^3 + 1 = y \Rightarrow x = \sqrt[3]{y-1}$. For $y \in \mathbb{R}$, $\sqrt[3]{y-1}$ is also in $\mathbb{R}$. ✓

Since f is 1-1 and onto, so it is invertible.

Sequences

A sequence on $\mathbb{N}$ is a list $s_0, s_1, \dots, s_n, \dots$ that has a specified value s_n for each integer $n \in \mathbb{N}$. s_n is the nth term of the sequence. The sequence itself is denoted by $s(n)$ or $(s_n)_{n\in\mathbb{N}}$ or $(s_0, s_1, s_2, \dots)$

Ex: The sequence $(s_n)_{n\in\mathbb{N}}$ where $s_n = n!$ is just the sequence $(1, 1, 2, 6, 24, 120, \dots)$. The set of values is $\{1, 2, 6, 24, 120, \dots\}$.

Ex: The sequence $(a_n)_{n\in\mathbb{N}}$ given by $a_n = (-1)^n$ for $n \in \mathbb{N}$ is the sequence $(1, -1, 1, -1, \dots)$ whose set of values is $\{-1, 1\}$

Ex: The following sequence have values that are sets.
A sequence $(D_n)_{n\in\mathbb{N}}$ of subsets of $\mathbb{Z}$ is defined by

$$D_n = \{m \in \mathbb{Z} : m \text{ is a multiple of } n\}.$$

Then
$$D_0 = \{0\}$$
$$D_1 = \mathbb{Z}$$
$$D_2 = \{\dots, -4, -2, 0, 2, 4, \dots\}.$$
$$\vdots$$

In general $D_n = \{0, \pm n, \pm 2n, \pm 3n, \dots\}$.

Ex: Let Σ be an alphabet. For each $k \in \mathbb{N}$, Σ^k is defined to be the set of all words in Σ^* having length k. In symbols

$$\Sigma^k = \{w \in \Sigma^* : \text{length}(w) = k\}.$$

The sequence $(\Sigma^k)_{k\in\mathbb{N}}$ is a sequence of subsets of Σ^* whose union $\bigcup_{k\in\mathbb{N}} \Sigma^k$, is Σ^*.

Note that Σ^k are disjoint.

For $\Sigma = \{a, b\}$

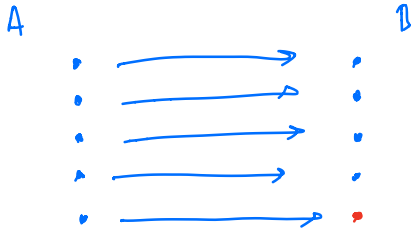
$\Sigma^0 = \{\lambda\}$, $\Sigma^1 = \{a, b\}$, $\Sigma^2 = \{aa, ab, ba, bb\}$.

$\Sigma^3 = \dots$.

Infinite sets

Def: The cardinality of a set A is equal to the cardinality of a set B, denoted $|A| = |B|$ iff there is a one-to-one correspondence from A to B.

A and B are of the same size.



Def: Any infinite set that is the same size the set $\mathbb{P}(\mathbb{Z}^+)$ of positive integers will be called countably infinite.

Thus, a set S is countably infinite if and only if there exists a one-to-one correspondence between $\mathbb{Z}^+$ and S.

Ex: The set $\mathbb{N}$ is countably infinite. Because $f(n) = n-1$ defines a 1-1 correspondence function mapping $\mathbb{Z}^+$ onto $\mathbb{N}$. $f(1)=0$, $f(2)=1$, ---

$f: \mathbb{Z}^+ \to \mathbb{N}$

Even though $\mathbb{Z}^+$ is a proper subset of $\mathbb{N}$, by definition $\mathbb{Z}^+$ is the same size as $\mathbb{N}$.

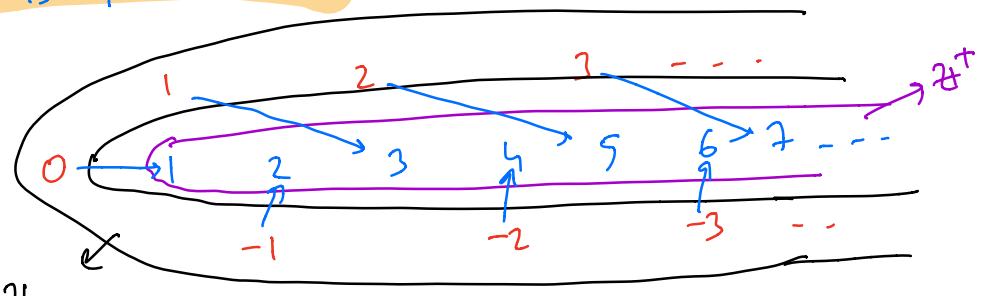
$\mathbb{Z}^+ \subset \mathbb{N}$

Ex: Is the set $\mathbb{Z}$ countable? $f: \mathbb{Z} \to \mathbb{Z}^+$?

Yes!

$$f(n) = \begin{cases} 2n+1 & \text{for } n \geq 0 \\ -2n & \text{for } n < 0 \end{cases}, \quad f: \mathbb{Z} \to \mathbb{Z}^+$$

$f(n)$ is 1-1 and onto. ✓



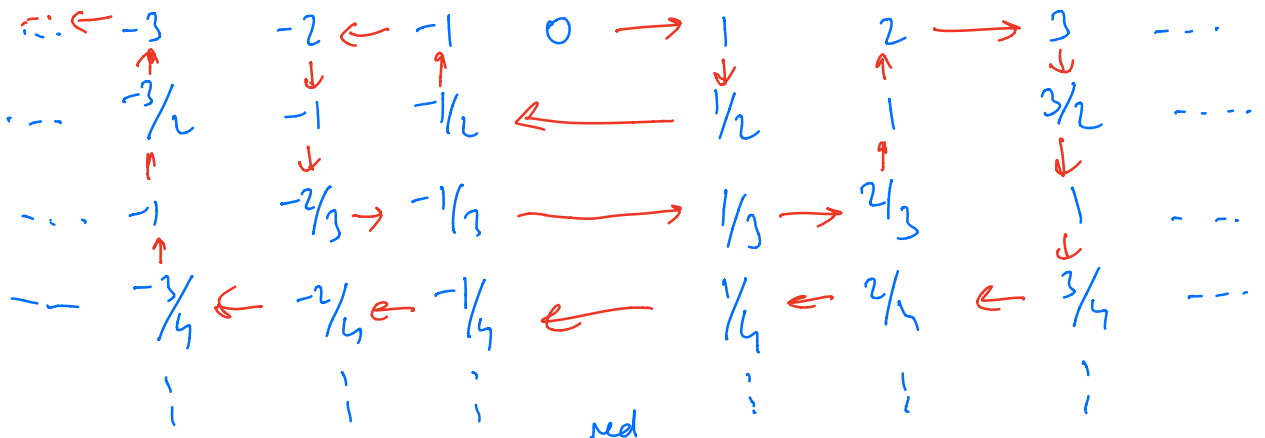
$\mathbb{Z}$

Therefore, $\mathbb{Z}$ is the same size as $\mathbb{Z}^+$.

$\mathbb{Z}$ is countable.

Ex: Is $\mathbb{Q}$ countable? $\mathbb{Q} = \{ x \mid x = \frac{a}{b}, a, b \in \mathbb{Z}, b \neq 0 \}$

Yes!



Obtain f by following the red arrows and skipping over repetitions.

Ex: Is $[0,1]$ countable? $\mathbb{Q} \subset \mathbb{R}$

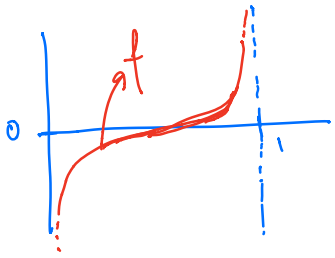
No!

Ex: Is $\mathbb{R}$ countable?

No!

Ex: Show that $\mathbb{R}$ and $(0,1)$ are in 1-1 correspondence!

$$f(x) = \frac{2x-1}{x\cdot(1-x)}$$



November 25

Theorem: The union of countably many countable sets is countable.

Ex: Show that if Σ is a finite alphabet, the set Σ^* of all words using letters from Σ is countably infinite.

Let Σ^k is defined to be the set of all words in Σ^* having length k where length(w) is the number of letters from Σ in w for $w \in \Sigma^*$.

Then, $\Sigma^k = \{w \in \Sigma^* : \text{length}(w) = k\}$

See that $\bigcup_{k \in \mathbb{N}} \Sigma^k = \Sigma^*$ (Note that Σ^k's are disjoint.)

Each Σ^k is finite i.e. countable and number of Σ^k's is countable. Therefore, by the above theorem, Σ^* is countable.

Ex: Show that $\mathbb{Q}$ is countable.

Remember $\mathbb{Z}$ and $\mathbb{Z}_i$ as $\mathbb{Z}/i$, $i \in \mathbb{Z}^+$. Then,

$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$

$\mathbb{Z}_1 = \mathbb{Z}$

$\mathbb{Z}_2 = \{\ldots, -3/2, -2/2, -1/2, 0, 1/2, 2/2, 3/2, \ldots\}$

$\mathbb{Z}_3 = \{\ldots, -3/3, -2/3, -1/3, 0, 1/3, 2/3, 3/3, \ldots\}$

$\vdots$

Clearly $\bigcup_{i \in \mathbb{Z}^+} \mathbb{Z}_i = \mathbb{Q}$

Since each $\mathbb{Z}_i$ is countable and there are countable $\mathbb{Z}_i$'s, their union is also countable, by the above theorem.